Mathematical Modeling

Lecture 1: Modeling Optimization Problems

Personal study notes

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1 What mathematical programming means

Here, *mathematical programming* means mathematical optimization: selecting the best feasible decision according to a specified criterion. The word "programming" refers to planning; an optimization problem is a mathematical object even before any computer program is written.

A model has three central ingredients:

1. **Decision variables**: quantities that may be chosen.
2. **Constraints**: conditions that an acceptable decision must satisfy.
3. **Objective function**: a numerical criterion to minimize or maximize.

Parameters, such as available resources or prices, are fixed data for one model instance. A parameter can change between instances without becoming a decision variable.

2 The general constrained problem

Let $x = (x_1, \dots, x_n)^T \in \mathbb{R}^n$ be the decision vector. A constrained minimization problem is

$$\min_{x \in \Omega} f(x), \qquad \Omega = \{x \in \mathbb{R}^n : g_i(x) \le 0 \ (i \in I),\ h_j(x) = 0 \ (j \in E)\}. \tag{1}$$

Here f is the objective, g_i are inequality-constraint functions, and h_j are equality-constraint functions; I and E are finite index sets. Bounds and sign restrictions must also be included in Ω when required.

Definition 2.1 (Feasibility and global optimality)**.** A point x is *feasible* if $x \in \Omega$. A feasible point x^* is a *global minimizer* if $f(x^*) \leq f(x)$ for every $x \in \Omega$. The value $f(x^*)$ is the optimal objective value.

An empty feasible set means that the requirements cannot be met simultaneously. Feasibility alone does not imply optimality. The optimal decision x^* and its objective value $f(x^*)$ are different objects, and multiple decisions may have the same value.

Remark 2.2 (Minimum versus infimum)**.** A finite lower bound need not be attained in a general optimization problem. For example, $\inf_{x>0} x = 0$, but no feasible point has value zero. A solver's output therefore requires both feasibility and an appropriate optimality or attainment argument.

3 From a real question to a mathematical model

1. **Choose variables and construct the model.** What can actually be controlled? What is given? State each variable's meaning, domain, and unit.
2. **Specify constraints.** Translate capacities, balances, required levels, and physical limits into equations or inequalities. Check their direction.
3. **Specify the objective.** State what "best" means, whether the criterion is minimized or maximized, and which units measure it.
4. **Select an algorithm.** Match the method to the model's structure: linearity, smoothness, convexity, dimensions, and continuous or discrete variables.
5. **Find and verify a solution.** Check feasibility and theoretical optimality conditions, then interpret the answer in the original setting.
6. **Perform post-optimal analysis.** Investigate how the answer depends on data and assumptions.

A correct solution to an inaccurate model can still be a poor real-world decision.

3.1 Units and assumptions

In a resource constraint, a coefficient measured in kg/item times a variable measured in items produces kg; its right-hand side must also be kg. Price in PLN/item times quantity in items produces revenue in PLN.

State whether quantities may be fractional. A count of indivisible objects requires integrality; permitting real values gives a continuous relaxation. State assumptions such as fixed coefficients, additive contributions, and unchanging prices. These are claims about the setting, not facts guaranteed by an optimization algorithm.

4 How optimization problems are classified

Constraints
: Unconstrained or constrained optimization.

Variables over time
: Static decisions or decisions evolving over time. Dynamic problems include control and functional optimization; this is broader than the algorithmic technique called dynamic programming.

Number of objectives
: Single-objective or multiobjective optimization. Several competing criteria require an explicit way to compare trade-offs.

Function structure
: Linear, nonlinear, quadratic, or convex optimization. Smoothness is another distinction; a nonlinear function may be smooth or nonsmooth.

Allowed values
: Continuous, discrete, integer, or binary variables.

A linear program has a linear objective, linear constraints, and continuous variables. Adding integrality produces an integer linear program. Quadratic programming does not automatically mean convexity. For $f(x) = \frac{1}{2}x^T Q x + c^T x$ with symmetric Q, convexity requires Q to be positive semidefinite; a convex optimization problem also needs a convex feasible set.

5 Optimization roadmap

1. **Linear programming**: modeling examples, standard and canonical forms, graphical solutions, bases, simplex methods, duality, post-optimal analysis, and interior-point methods.
2. **Unconstrained nonlinear optimization**: local optimality conditions, iterative methods, convergence, line searches, steepest descent, Newton and quasi-Newton methods, conjugate gradients, and direct search.
3. **Constrained nonlinear optimization**: Lagrange multipliers, Karush–Kuhn–Tucker conditions, quadratic and convex problems, penalty and barrier methods, and interior-point methods.

The simplex method for linear programming and the Nelder–Mead simplex method are different algorithms. Their shared name does not make them interchangeable.

6 MATLAB refresher scope

Core MATLAB skills include vectors and matrices, programming constructs, built-in functions (including numerical methods), scripts, function files, parameter passing (including passing functions), and basic plots. Optimization Toolbox provides solvers for comparison with implemented algorithms.

7 Questions for revision

1. Which quantities in a model are decisions and which are parameters?
2. What distinguishes a feasible point, an optimal point, and an optimal value?
3. How would changing an inequality's direction change the model?
4. Why can fractional counts be mathematically feasible but physically unusable?
5. What must be checked after an algorithm returns a candidate solution?

8 Sources and editorial remarks

Primary source: `s4_w1.pdf`, PDF pages 1–6.
Source context: Folder labelled 2020; the slides contain a 2019/2020 schedule. Exact lecture date is not established.
Current course alignment: Unconfirmed.
Editorial scope: English synthesis of the mathematical content. Definitions and explanations expand the brief slides; historical schedules, assessment rules, and contact details are not current course instructions.

Polish equivalents are maintained in `docs/glossary.md`, including *zmienna decyzyjna*, *funkcja celu*, *ograniczenie*, *rozwiązanie dopuszczalne*, and *rozwiązanie optymalne*.

- `s4_w1.pdf`, pp. 1–2: historical administration and bibliography; not reproduced as current instructions.
- pp. 3–4: Sections 1–4 and the start of Section 5.
- pp. 5–6: remaining roadmap in Section 5 and MATLAB scope in Section 6.

Page numbers refer to PDF pages, each containing up to four slides.